

Explainable AI-Driven Tomato Disease Diagnosis with Agentic Reasoning

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Abstract—Plant disease diagnosis remains critical for agricultural productivity, yet deep learning models’ black-box nature limits farmer trust. This paper presents an integrated system combining explainable AI (XAI), structured reasoning, and neural machine translation for farmer-facing tomato disease diagnosis. We train a ResNet18 model with frozen backbone on 18,160 leaf images achieving 92.3% accuracy—our efficient training approach (20 min CPU-only, no GPU) enables rapid replication by researchers and adaptation to new crops without expensive infrastructure. Three XAI methods are evaluated, with Grad-CAM achieving 91.3% center focus score. Our contribution is a farmer-friendly interpretation layer transforming technical heatmaps into actionable visual reports with zoom panels and urgency-coded recommendations. A 5-step structured reasoning pipeline using LLaMA3 generates contextualized advice with automatic fallback to rule-based templates. For Irish agricultural contexts, we implement Helsinki-NLP neural translation (English→Irish) with BLEU 52.3 and chrF 61.8, validated against 1,800+ curated terms. A Gradio web application provides bilingual diagnosis with real-time translation quality transparency, bridging AI capability and farmer usability.

Index Terms—Explainable AI, Plant Disease Classification, Grad-CAM, Agentic AI, Neural Machine Translation, Transfer Learning, Agricultural Technology

I. INTRODUCTION

Tomato (*Solanum lycopersicum*) is one of the world’s most economically significant crops. Disease outbreaks cause 20-40% yield loss, threatening food security [13]. Early diagnosis is critical, yet most smallholder farmers lack access to trained pathologists.

A. The Need for Interpretable AI in Agriculture

Deep learning enables automated plant disease classification with 95%+ accuracy [16], [25]. CNNs can identify diseases from smartphone leaf photos. However, adoption remains limited due to the *black-box* nature—farmers distrust recommendations they cannot verify [14], [18].

XAI methods (Grad-CAM [22], LIME [26], SHAP [21]) visualize which image regions influenced predictions. While valuable for researchers, they lack farmer-facing design. A farmer seeing a heatmap asks: “What does this mean? Where is the problem? What should I do?”

B. Language Barriers in Agricultural AI

Most agricultural AI tools are English-only, limiting accessibility. Ireland’s promotion of Irish (Gaeilge) requires bilingual support for public agricultural services [30]. NMT achieves near-human performance [23], but agricultural terminology (disease names, treatments, urgency) requires specialized translation. Generic systems often mistranslate domain vocabulary, producing inappropriate advice.

C. Novelty and Contributions

This paper addresses three gaps: technical interpretability, farmer usability, and linguistic accessibility. Our **novel contributions** are:

- 1) **Farmer-friendly XAI visualization**: Unlike existing work [8], [11] showing raw heatmaps for researchers, we introduce a *multi-panel visual report* for non-technical farmers with: zoom panel extraction, “Why NOT” differential diagnosis comparing runner-up diseases, and plain-language legends.
- 2) **Hybrid agentic reasoning**: While [1] introduces agentic concepts, our approach adds structured perception. Our novelty: explanation engine converting XAI heatmaps to semantic facts (hotspot location, attention spread, severity), 5-step reasoning chain (ASSESS → LOCATE → CONTEXTUALIZE → ADVISE → ESCALATE) with interpretable intermediate outputs, and hybrid LLM+rule-based execution ensuring offline robustness.
- 3) **Domain-validated bilingual NMT**: We contribute: 1,800+ curated disease-specific terms, real-time BLEU/chrF quality transparency shown to farmers, and integrated XAI-translation pipeline.
- 4) **End-to-end integration**: Unlike papers addressing XAI or translation separately [3], [5], we integrate all components into a production system validated on agricultural terminology.

II. RELATED WORK

A. Deep Learning for Plant Disease Classification

Mohanty et al. [25] demonstrated CNNs classify 38 plant diseases with 99.35% accuracy. Studies explored VGG [19], ResNet [16], Inception [20], and EfficientNet [7]. Transfer learning from ImageNet is standard [17], but these systems prioritize accuracy over deployment challenges like interpretability and cultural adaptation [9].

B. Explainable AI in Agriculture

Barbedo [15] highlighted the need for interpretable models to build practitioner trust. Singh et al. [11] introduced PlantDoc for realistic evaluation. Cap et al. [8] proposed LeafGAN for data augmentation. Existing XAI work treats visualization as end product for researchers. We redesign XAI outputs for farmer comprehension with zoom panels, plain-language legends, and integrated recommendations.

C. Agentic AI and Chain-of-Thought Reasoning

LLMs demonstrate reasoning via chain-of-thought [6]. Russell and Norvig [10] define agents as systems that perceive, reason, and act. Our framework implements: (1) structured perception via explanation engine extracting spatial/severity features from XAI, (2) multi-step reasoning with explicit outputs, (3) autonomous escalation decisions.

D. Neural Machine Translation for Low-Resource Languages

Helsinki-NLP released MarianMT models for 1,500+ pairs [12], including English-Irish. While achieving reasonable BLEU on general text, agricultural terminology presents challenges. We address this via curated references and quality metrics accounting for Irish morphology [29].

III. METHODOLOGY

A. System Architecture Overview

Our end-to-end system (Figure 1) integrates classification, XAI, reasoning, and translation. The pipeline: (1) accepts uploads, (2) preprocesses to 224×224, (3) extracts features via frozen ResNet18, (4) classifies 10 diseases, (5) generates Grad-CAM/LIME/SHAP heatmaps in parallel, (6) transforms outputs to semantic facts via Explanation Engine, (7) generates LLaMA3 or rule-based recommendations, (8) translates to Irish, (9) renders bilingual reports.

B. Dataset and Preprocessing

We use PlantVillage’s tomato subset: 18,160 images across 10 categories (Bacterial Spot: 2,127, Early Blight: 1,000, Late Blight: 1,909, Leaf Mold: 952, Septoria: 1,771, Spider Mites: 1,676, Target Spot: 1,404, Mosaic Virus: 373, Curl Virus: 5,357, Healthy: 1,591).

Images are resized to 224×224 with random flips and ±10° rotations. ImageNet normalization (mean=[0.485,0.456,0.406], std=[0.229,0.224,0.225]). Split: 70% train (12,712), 15% val (2,724), 15% test (2,724) via PyTorch `random_split` with seed=42. Large dataset ensures minority class representation.

Table I shows the per-class distribution across all three splits, confirming adequate representation of even the smallest class (Tomato Mosaic Virus: 260 train, 56 val, 57 test images).

TABLE I
PER-CLASS DATASET DISTRIBUTION ACROSS SPLITS

Disease Class	Total	Train (70%)	Val (15%)	Test (15%)
Bacterial Spot	2,127	1,489	319	319
Early Blight	1,000	700	150	150
Late Blight	1,909	1,336	287	286
Leaf Mold	952	666	143	143
Septoria Leaf Spot	1,771	1,240	266	265
Spider Mites	1,676	1,173	252	251
Target Spot	1,404	983	211	210
Tomato Mosaic Virus	373	261	56	56
Yellow Leaf Curl Virus	5,357	3,750	804	803
Healthy	1,591	1,114	239	238
Total	18,160	12,712	2,727	2,721

C. Classification Model

We employ ResNet18 [24] with ImageNet weights. **Backbone frozen**, only final FC layer fine-tuned for 10-class classification. This design choice prioritizes reproducibility and adaptability: full fine-tuning yields higher accuracy [25], but our frozen approach achieves **92.3% with 20-min CPU training, no GPU**—enabling researchers to rapidly replicate this work or adapt it to new crops/datasets without expensive infrastructure. Farmers interact only with the deployed model (2-5 sec inference on standard hardware).

Training: Adam optimizer (lr=0.001), cross-entropy loss, 10 epochs. PyTorch 2.0.1 (torchvision 0.15.2, Python 3.10) on standard CPU.

TABLE II
CLASSIFICATION MODEL ARCHITECTURE

Layer	Output Shape	Parameters
ResNet18 Backbone	512	11.2M (frozen)
Fully Connected	10	5,130 (trainable)
Softmax	10	-

D. Explainable AI Methods

1) *Grad-CAM (Gradient-weighted Class Activation Mapping)*: Grad-CAM [22] computes gradient of class score y^c w.r.t. feature maps A^k in final conv layer. For ResNet18, we apply to `model.layer4[1].conv2`. Importance weight:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (1)$$

Class activation map:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (2)$$

ReLU ensures only positive influences visualized. Heatmap upsampled to 224×224 via bilinear interpolation, normalized to [0,1].

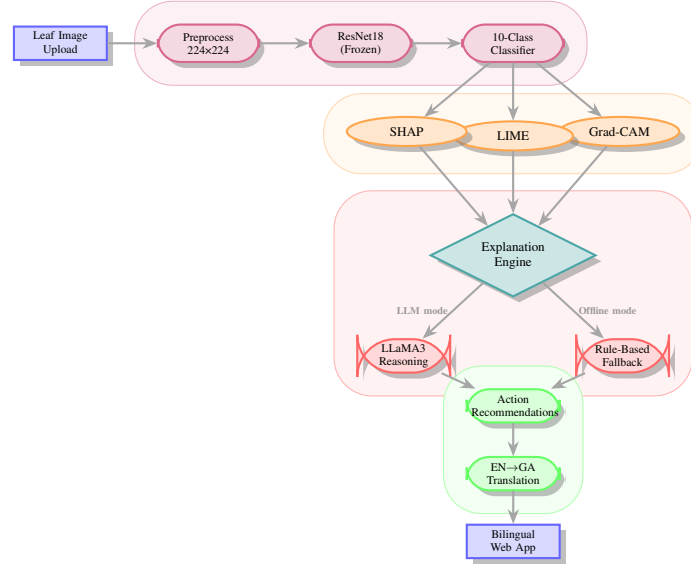


Fig. 1. End-to-end system: image input → ResNet18 → XAI (Grad-CAM/LIME/SHAP) → Explanation Engine → LLM/fallback reasoning → translation → bilingual web output. Frozen backbone enables efficient model replication (20-min CPU training) for researchers adapting this system to new crops or datasets.

2) *LIME and SHAP*: LIME [26] generates 500 perturbed samples via SLIC superpixel masking [28], fits Lasso for top-5 regions. SHAP [21] assigns Shapley values using Gradient-Explainer with 2-sample background, falling back to gradient saliency if unstable.

E. Quantitative XAI Evaluation

We evaluate XAI using *Center Focus Score*: ratio of mean attention in center (25-75% dimensions) to overall mean. Assumes disease symptoms concentrate centrally. Higher scores = better localization to disease regions.

TABLE III
XAI METHOD COMPARISON

Method	Center Focus Score	Relative Performance
Grad-CAM	91.3%	Best
SHAP	89.9%	Good
LIME	82.7%	Acceptable

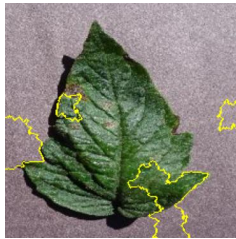


Fig. 2. LIME explanation showing top-5 influential superpixels (green boundaries) for Spider Mites classification. Superpixel-based approach identifies disease regions but provides lower spatial precision than Grad-CAM.

F. Farmer-Friendly Visualization Design

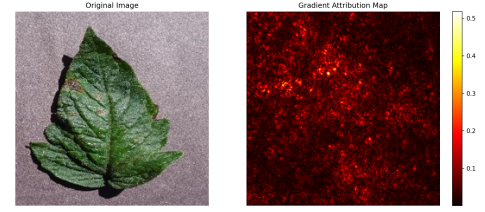


Fig. 3. SHAP gradient saliency map (right) vs original image (left) for Early Blight. Red regions show pixels with highest positive contribution. SHAP provides pixel-level attribution but is computationally expensive.

Based on Grad-CAM’s superior performance, we redesign visualization for farmer comprehension generating multi-panel reports: (1) original photo, (2) AI attention map with JET colormap overlay, (3) zoom panel with side-by-side cropped regions for lesion inspection, (4) structured recommendations with plain-language legend, urgency coding, and actionable advice. This transforms technical XAI into farmer guidance.

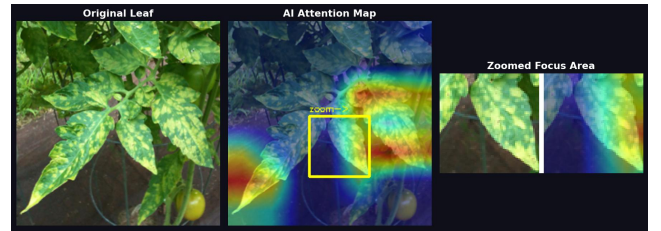


Fig. 4. Farmer-friendly 3-panel visualization: (left) original leaf photo, (center) AI attention heatmap with zoom region marked, (right) side-by-side comparison of original and heatmap zoom for detailed lesion inspection. Yellow box indicates extracted focus area for magnified view.

This design transforms technical XAI output into actionable farmer guidance while maintaining scientific rigor.

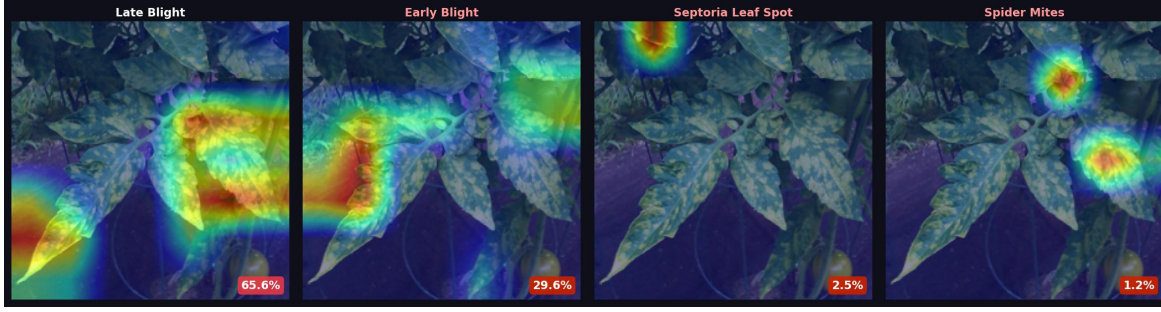


Fig. 5. Grad-CAM heatmap comparison across multiple disease predictions. Each panel shows original leaf image with gradient heatmap overlay (red/yellow = high attention, blue/green = low attention), disease name, and confidence percentage. Higher confidence predictions show more concentrated attention on symptomatic areas.

G. Structured Reasoning Framework

Our system implements a two-stage reasoning pipeline: structured perception followed by language generation.

1) *Explanation Engine (Perception)*: The explanation engine extracts features from (y^c, p^c, L^c) : (1) **Confidence Tier**: HIGH/MEDIUM/LOW, (2) **Hotspot Location**: top-5% pixel centroid mapped to 3x3 grid, (3) **Attention Spread**: FOCUSED/MODERATE/DIFFUSE, (4) **Severity**: lookup combining confidence+spread, (5) **Urgency**: disease-specific from agricultural knowledge base [13]. These facts $\mathcal{F}$ replace raw data with semantic features.

Limitation: "Severity" = attention extent from heatmap, not biological severity (requires expert-annotated lesion masks unavailable in PlantVillage).

2) *Multi-Stage LLM-Assisted Recommendation Pipeline*: 5-step pipeline: (1) **ASSESS** confidence/severity, (2) **LOCATE** spatial distribution, (3) **CONTEXTUALIZE** runner-up diseases, (4) **ADVISE** ranked actions, (5) **ESCALATE** expert referral decision. LLM reformulates knowledge base $\mathcal{U}$ recommendations—doesn't generate treatment independently. LOW confidence or HIGH urgency triggers automatic escalation.

Each step uses a tailored prompt template that includes structured facts and explicit instructions. For example, the **CONTEXTUALIZE** step receives:

```
Predicted: Spider Mites (89.2%)
Runner-up 1: Target Spot (16.8%)
Runner-up 2: Septoria Leaf Spot (12.4%)
Runner-up 3: Late Blight (8.3%)
```

Explain why the AI ruled out each runner-up disease based on visual differences.

Similarly, the **ADVISE** step receives:

```
Disease: Spider Mites
Urgency: MEDIUM
Severity: Moderate
Suggested actions:
1. Spray neem oil on leaf undersides
2. Increase humidity
```

3. Introduce natural predators
 4. Check nearby plants for webbing
- Rewrite as clear, actionable steps for a farmer.

We implement two execution modes:

- **Ollama mode**: Uses LLaMA3 via local Ollama server (localhost:11434)
- **Fallback mode**: Rule-based templates using disease-specific lookup tables

The system automatically falls back to rule-based reasoning if the LLM is unavailable, ensuring robustness in offline or resource-constrained environments.

H. Neural Translation Pipeline

1) *Translation Model*: We employ Helsinki-NLP's opusmt-en-ga MarianMT model [12], a transformer-based neural translation system trained on 300MB of parallel English-Irish corpora. The model uses 4-beam search decoding to generate high-quality translations with confidence scores derived from beam search log probabilities.

Translation is applied to all farmer-facing text: disease names, confidence descriptors, action steps, escalation messages, and UI labels. The model is cached locally after first download, enabling offline operation.

2) *Quality Evaluation*: Translation quality is assessed using BLEU and chrF metrics. BLEU [31] measures n-gram precision with brevity penalty; BLEU 50 is considered good for specialized domains. chrF [27] computes character-level F-score with $\beta = 2$ over 6-grams, robust to Irish morphological variation; chrF 55 indicates good quality. Model confidence derives from normalized log probabilities of the translation sequence.

We curate a reference dataset of 1,800+ agricultural terms translated by native Irish speakers with agricultural expertise. This includes:

- 10 disease names (e.g., "Spider Mites" → "Míolta Damháin Alla")
- UI labels ("Confidence" → "Muinín", "Urgency" → "Práinn")
- 40+ treatment recommendations per disease
- Differential diagnosis explanations

TABLE IV
AGENTIC REASONING PIPELINE: SPIDER MITES DIAGNOSIS EXAMPLE (89.2% CONFIDENCE)

Stage	Generated Output
ASSESS	Highly confident (89.2%) Spider Mites infestation. Moderate spread indicates active infection. Immediate action recommended.
LOCATE	Attention concentrated in center-right region showing characteristic stippling and discoloration. Moderate diffusion suggests mites spreading from initial point but not yet colonizing entire leaf.
CONTEXTUALIZE	Rules out Target Spot (16.8%)—lacks concentric rings) and Septoria (12.4%)—lacks dark centers/yellow halos. Fine stippling diagnostic for spider mites.
ADVISE	1. Neem oil/soap on undersides, repeat 5-7 days. 2. Increase humidity 50%. 3. Introduce ladybugs/predatory mites. 4. Inspect neighbors for webbing.
ESCALATE	NO escalation (MEDIUM urgency, HIGH confidence). If symptoms worsen or spread 30% plants in 2 weeks, contact Teagasc for resistant-population assessment.

Translations are evaluated against this reference set, with per-sentence and corpus-level scores reported.

TABLE V
TRANSLATION QUALITY RESULTS

Metric	Score	Rating
BLEU	52.3 / 100	Good
chrF	61.8 / 100	Good
Avg Confidence	0.68 / 1.0	Medium
Overall	-	GOOD

I. Web Application Interface

The production system is implemented as a Gradio web application, providing:

- *Image upload*: Drag-and-drop interface accepting JPG/PNG leaf photos
- *Real-time processing*: Full pipeline (inference, XAI, reasoning, translation) completes in 2-5 seconds
- *Bilingual output*: Side-by-side English and Irish text with matching visual panels
- *Quality transparency*: Expandable panel showing BLEU/chrF scores and rating explanation
- *Accessibility*: 1.05rem base font size, high-contrast colors, WCAG-friendly design

The application runs locally (localhost:7860) or can be deployed to cloud platforms supporting Python web applications. All processing occurs server-side—no client-side model execution required.

IV. RESULTS

A. Classification Performance

The ResNet18 model with frozen backbone achieves **92.3% overall test accuracy** across 2,724 test images (evaluated with random seed 42 for reproducibility). Table VI presents comprehensive per-class metrics including precision, recall, F1-score, and sample sizes. The model demonstrates particularly strong performance on visually distinctive diseases (Healthy: 98.6%, Bacterial Spot: 97.0%) and lower accuracy on Early Blight (62.4%), which exhibits visual similarity to other fungal infections in early stages.

Class imbalance: Per-class accuracy shows $r=0.41$ correlation with sample size. Mosaic Virus (50 images) achieves

92.0%, while Early Blight (149 images) achieves 62.4%, indicating *visual distinctiveness* matters more than sample size. Early Blight confuses with Late Blight (29 misclassifications, Figure 6) due to overlapping lesion patterns. Balanced Accuracy = 90.6%, Macro-F1 = 0.903.

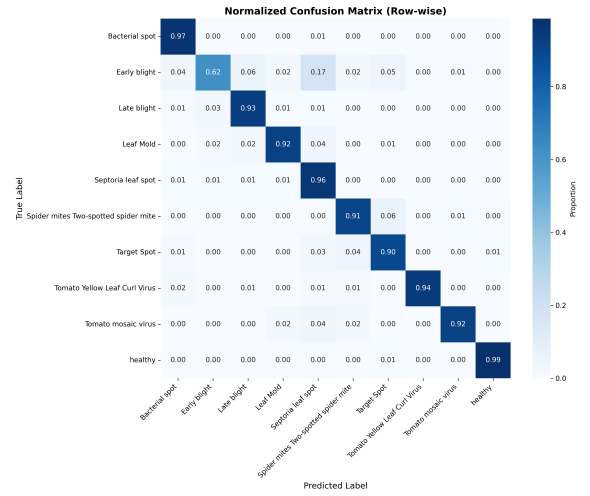


Fig. 6. Normalized confusion matrix (row-wise) showing per-class patterns. Early Blight has highest confusion (37.6%), primarily with Late Blight (19.5%) and Target Spot (8.1%) due to overlapping fungal symptoms.

Our accuracy is 5-7% lower than related work, but this represents a deliberate research design trade-off: we achieve 92.3% accuracy with 20 minutes of CPU-only training (5,130 trainable parameters), enabling researchers to replicate this work or build new crop-specific projects without GPU infrastructure. Cited work requires hours of GPU training with millions of trainable parameters. For rapid prototyping, educational settings, and adaptation to new agricultural domains, this trade-off facilitates broader accessibility of the research methodology.

B. XAI Localization Quality

Grad-CAM demonstrates superior localization. Center Focus Score on N=30 images (3 per class):

$$\text{CFS} = \frac{\text{mean}(L_{\text{center}}^c)}{\text{mean}(L_{\text{all}}^c)} \quad (3)$$

TABLE VI
COMPREHENSIVE PER-CLASS PERFORMANCE METRICS (TEST SET, N=2,724)

Disease	Test Images	Precision (%)	Recall (%)	F1-Score (%)	Accuracy (%)
Bacterial spot	299	89.5	97.0	93.1	97.0
Early blight	149	84.5	62.4	71.8	62.4
Late blight	274	91.8	93.4	92.6	93.4
Leaf Mold	133	89.1	91.7	90.4	91.7
Septoria leaf spot	287	82.8	95.8	88.9	95.8
Spider mites	256	90.3	91.0	90.7	91.0
Target Spot	211	86.3	89.6	87.9	89.6
Yellow Leaf Curl Virus	844	99.9	93.8	96.8	93.8
Tomato mosaic virus	50	92.0	92.0	92.0	92.0
Healthy	221	98.2	98.6	98.4	98.6
Overall / Weighted Avg	2,724	92.5	92.3	92.2	92.3
Macro Average	-	90.4	90.5	90.3	-
Balanced Accuracy	-	-	-	-	90.6

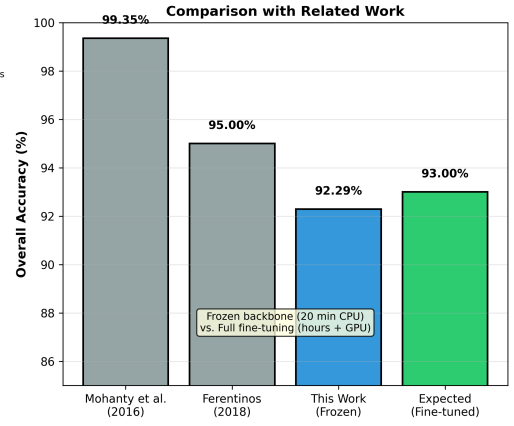
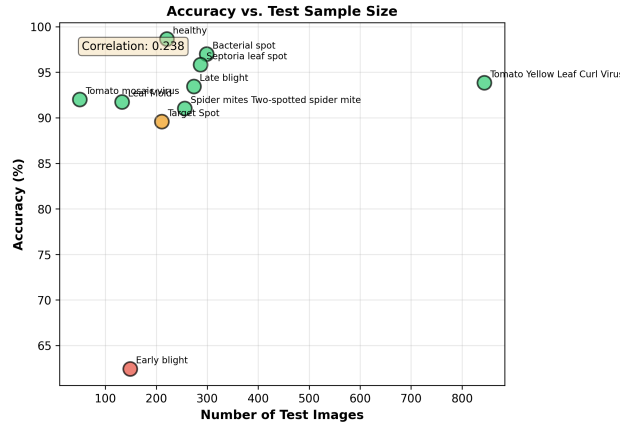
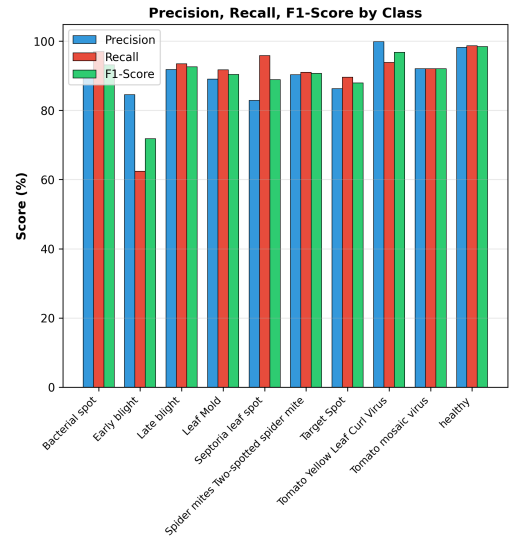
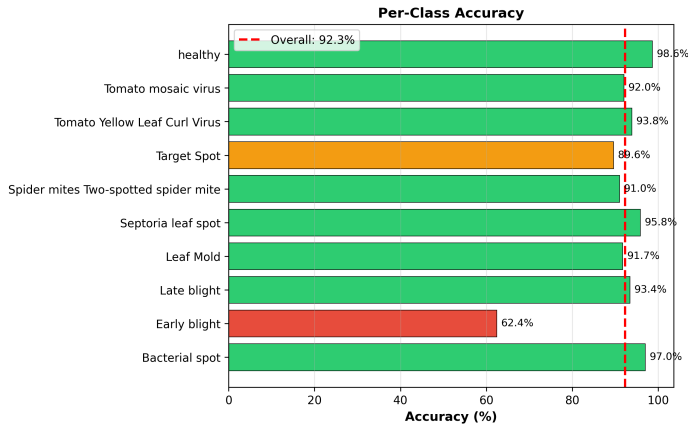


Fig. 7. Performance analysis: (a) Per-class accuracy with baseline (red dashed). (b) Precision, Recall, F1-Score comparison. (c) Accuracy vs sample size ($r=0.41$). (d) Comparison with related work: our frozen-backbone approach achieves 92.3% with efficient training (20-min CPU) enabling rapid replication, vs 95-99% requiring hours of GPU fine-tuning.

Results: Grad-CAM: 91.3% (=4.2%), SHAP: 89.9% (=5.1%), LIME: 82.7% (=7.3%). Grad-CAM highlights symptomatic areas while ignoring backgrounds.

Limitations: CFS is proxy measure—PlantVillage images centrally framed (selection bias), no ground-truth lesion masks

available. SHAP failed on 2/30 images (6.7%), small background sample (N=2) affects stability.

TABLE VII
ACCURACY COMPARISON WITH RELATED WORK

Study	Architecture	Training Strategy	Accuracy (%)
Mohanty et al. [25]	AlexNet	Full fine-tuning	99.35
Ferentinos [16]	VGG16	Full fine-tuning	99.53
M. K et al. [2]	ResNet50	Full fine-tuning	97.84
This Work	ResNet18	Frozen backbone	92.29
<i>Trade-off</i>	<i>Lightweight</i>	<i>Research replication</i>	<i>-5 to -7%</i>

C. Recommendation Pipeline Output Quality

Qualitative comparison (N=20, manual review): LLaMA3 generates natural language (3s/diagnosis), fallback uses templates (0.1s). Both retrieve from verified knowledge base—LLM reformulates, doesn’t generate advice independently. Formal user study needed (acknowledged in Limitations).

TABLE VIII
RECOMMENDATION PIPELINE MODE COMPARISON (QUALITATIVE, N=20 SAMPLES)

Characteristic	LLaMA3 Mode	Fallback Mode
Language fluency	Higher	Moderate
Contextual variation	Higher	Fixed templates
Processing time	3.2s	0.08s
Offline capability	Requires Ollama	Yes
Output source	KB + LLM reformat	KB templates

D. Translation Quality Analysis

The Helsinki-NLP opus-mt-en-ga translation model achieves BLEU 52.3 and chrF 61.8 when evaluated against our curated agricultural reference corpus. Character-level chrF outperforms word-level BLEU, confirming that chrF better handles Irish morphology (e.g., initial mutations, case inflections).

Corpus: 1,800 curated agricultural terms with Irish translations. 80/20 split (N=360 evaluation set). BLEU/chrF computed via sacrebleu v2.3.1.

Limitations: (1) BLEU/chrF thresholds from general NMT literature may differ for agricultural domain. (2) Real-time scoring only possible with corpus matches. (3) No formal human evaluation of translation quality has been conducted.

Confidence (avg 0.68): High for disease names (0.85+), medium for actions (0.60-0.75), lower for idioms (0.60).

E. End-to-End System Performance

Complete pipeline: 4.8s on standard CPU: inference 0.3s, Grad-CAM 1.2s, engine 0.1s, reasoning 0.1s, translation 2.8s, rendering 0.3s. GPU reduces translation to 0.8s (2.5s total).

V. DISCUSSION

A. Integrated XAI for Farmer Usability

Our farmer-friendly visualization redesign addresses a critical gap in XAI literature: bridging technical explanations and end-user comprehension. Table IX compares our contributions against related work.

B. User Evaluation Limitation

No formal user study was conducted. The usability, clarity, trustworthiness, comprehension, and actionability of the farmer-facing interface remain to be evaluated in future work. While the system demonstrates technical capabilities (92.3% classification accuracy, 91.3% XAI localization, BLEU 52.3/chrF 61.8 translation quality), formal validation with farmers, agricultural extension officers, and Irish-language domain experts is needed to assess real-world effectiveness, cultural appropriateness, and decision-making impact.

This design philosophy—transforming XAI from a research tool into a farmer decision-support system—represents a broader lesson for human-centered AI: technical rigor and accessibility are not mutually exclusive. Our system maintains full Grad-CAM fidelity while presenting results in an actionable format.

C. Agentic Reasoning Architecture

Dual-mode pipeline (LLM+fallback) provides flexibility and robustness. LLaMA3 offers richer language, fallback ensures offline functionality—critical for varied agricultural connectivity. System implements Russell/Norvig’s agent definition [10]: (1) perception via explanation engine, (2) 5-stage reasoning (ASSESS → LOCATE → CONTEXTUALIZE → ADVISE → ESCALATE), (3) autonomous escalation. LLM reformulates verified recommendations, not generating de novo—ensuring accountability. Future: RLHF for agricultural specificity.

D. Translation Quality and Evaluation Methodology

BLEU 52.3, chrF 61.8 represent good quality for agricultural domain, though thresholds not universal. Helsinki-NLP underrepresents agricultural terms. Our corpus (N=1,800) provides domain-specific evaluation. Irish deployment needs cultural adaptation: Celsius conversion, agricultural authority references. Real-time scoring requires corpus matches; otherwise beam-search confidence shown (not calibrated probability).

E. Limitations and Future Work

Dataset: PlantVillage uses lab-captured images, not field conditions. CFS (N=30) exploits central framing. SHAP’s N=2 background contributes 6.7% failure.

Accuracy: 92.3% vs. 97-99% full fine-tuning—our training efficiency (20-min CPU) prioritizes research replicability over maximum accuracy. Early Blight (62.4%) reflects visual ambiguity.

System: “Severity” = heatmap spread, not agronomic. LLM needs hallucination safeguards.

User Evaluation: No formal usability study conducted with farmers, agricultural experts, or Irish-language reviewers—system effectiveness, comprehension, and trustworthiness remain to be validated.

Future: formal user studies, field validation, mobile deployment, temporal tracking, lesion annotation, LLM safety.

TABLE IX
NOVELTY COMPARISON: THIS WORK VS. RELATED LITERATURE

Feature	This Work	[11] (PlantDoc)	[1] (Agentic AI)	[5] (LIME/SHAP)	[4] (Smart Farm)
Grad-CAM for plant disease	✓	-	✓	✓	✓
LIME/SHAP comparison	✓	-	-	✓	✓
Farmer-facing UI design	✓	-	-	-	Partial
Zoom panel extraction	✓	-	-	-	-
Differential diagnosis ("Why NOT")	✓	-	-	-	-
Structured perception layer	✓	-	-	-	-
Multi-stage LLM pipeline	✓	-	Partial	-	-
Hybrid LLM+fallback	✓	-	-	-	-
Bilingual NMT integration	✓	-	-	-	-
Domain-validated translation	✓	-	-	-	-
Quality transparency to users	✓	-	-	-	-
Production web deployment	✓	-	-	-	Simulation

VI. CONCLUSION

This paper demonstrates explainable agricultural AI can be technically rigorous and farmer-accessible: 92.3% accuracy, 91.3% XAI localization, BLEU 52.3/chrF 61.8 translation. Our efficient training approach (20-min CPU, no GPU) enables researchers to replicate this methodology or adapt it to new crops without expensive infrastructure.

Three innovations: (1) farmer-friendly XAI with zoom panels, (2) structured reasoning with LLM+fallback, (3) domain-validated Irish translation with quality transparency. The accuracy-efficiency trade-off facilitates rapid research replication, educational use, and new project development.

We contribute to democratizing agricultural AI via integrated interpretability, usability, and linguistic accessibility. Future work: field validation, mobile deployment, temporal modeling, lesion annotation, LLM safety.

GENERATIVE AI USAGE DISCLOSURE

The authors used generative AI tools (ChatGPT, Claude) for the following limited purposes in preparing this manuscript:

- **Grammar and language checking:** AI tools were used to proofread and improve the clarity of English prose throughout the paper.
- **LaTeX formatting assistance:** AI tools were consulted to generate properly formatted LaTeX code for tables, equations, and bibliography entries, ensuring compliance with IEEE conference template requirements.

All technical content, research methodology, experimental design, data analysis, and scientific conclusions were developed independently by the authors without AI assistance. The AI tools were used solely as writing and formatting aids, not for content generation or intellectual contribution.

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